A Systematic Approach to Intraday Forex Trading

Design, Backtesting, and Robustness Validation of a Multi-Timeframe Momentum and Structural Breakout Strategy

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Quantitative Trading System — Backtesting and Validation Pipeline

Abstract

This paper presents the design, implementation, and validation of a systematic intraday trend-following strategy applied to three major currency pairs — EURUSD, GBPUSD, and USDJPY — over the period January 2011 to April 2026. The strategy combines a multi-timeframe directional filter with short-term price-imbalance zones on the five-minute series as limit-order entry regions, and uses structure-referenced stop placement with a two-stage scaled exit. The underlying thesis is that the interaction of multi-timeframe directional persistence and intra-session liquidity-gap rebalancing creates a conditional structural edge in liquid FX pairs.

A Python-based backtesting pipeline evaluates performance under three execution-cost regimes, including a worst-case stress test at 2× estimated spread and slippage. Portfolio simulation models shared capital across all three instruments with continuous open-risk accounting and a 5.5% portfolio cap. Under realistic execution costs, the three-pair portfolio produced a total return of +136.71%, a maximum drawdown of -5.36%, and an annualized Sharpe ratio of 1.600 over the 15-year test period. Under worst-case execution, the strategy returned +62.97% with a maximum drawdown of -8.74% and a Sharpe ratio of 0.867.

Robustness testing — walk-forward validation, rolling window analysis, parameter sensitivity, and a 1,000-iteration bootstrap Monte Carlo — provides evidence that the observed performance is not attributable to overfitting or favorable trade sequencing. The approximate t-statistic on per-trade returns exceeds conventional significance thresholds, and all bootstrap resamples terminate above starting capital. An edge-classification framework identifies the strategy as a regime-conditional structural edge that is strongest during sustained directional market phases and weakest during macro-dominated or structurally disrupted periods. The results support a positive expected-value interpretation, while regime dependency and several execution-modeling constraints define the primary forward-looking risks.

1. Introduction

This paper documents the development of a fully systematic intraday trend-following strategy for spot forex. The work is motivated by the observation that directional continuations following multi-timeframe structural alignment in highly liquid currency pairs appear to be a potentially exploitable regularity, conditional on specific intra-session liquidity behavior. The goal was to translate that observation into a precise, rule-based system with quantifiable expected value, explicit execution-cost modeling, and a validation framework appropriate for assessing live deployment readiness.

The design requirements were fourfold. First, every entry, exit, and risk decision must be fully deterministic and rule-based, with no discretionary override. Second, execution costs must be modeled explicitly across multiple scenarios, including a stress case that substantially exceeds typical conditions. Third, performance must be validated on data not used in parameter selection, using both walk-forward and rolling window methods. Fourth, the portfolio simulation must reflect shared capital constraints rather than treating each instrument independently.

The result is an eight-module Python pipeline covering data ingestion, signal generation, trade simulation, cost modeling, analytics, walk-forward validation, sensitivity analysis, and portfolio simulation. The sections that follow cover the strategy's theoretical basis, the simulation architecture, robustness validation methodology, statistical significance testing, portfolio-level results under two cost scenarios, a benchmark comparison, an explicit limitations analysis, and an edge classification that characterizes the conditions under which the strategy is expected to perform.

2. Problem Statement

Systematic strategy development requires resolving three evaluation challenges before any performance figures can be considered credible.

The first is measurement. Without a fixed, rule-based signal definition, it is not possible to distinguish genuine positive expected value from selective observation under favorable conditions. Formalizing the strategy in code forces complete specification of every decision, which makes performance deterministic and auditable.

The second is reproducibility. A fully specified simulation engine applies identical logic to every bar in the historical series, which removes variation from changing market conditions or inconsistent execution. With a fixed random seed, results are reproducible across runs and environments.

The third is cost sensitivity. For an intraday strategy with modest per-trade R-multiples, spread and slippage represent a meaningful fraction of expected return. A strategy that appears profitable under zero-cost assumptions may become unprofitable once realistic friction is applied. Explicit cost modeling and multi-scenario testing are therefore required before any capital deployment.

This project addresses all three constraints by constructing a deterministic, fully logged simulation engine that operates on tick-approximate one-minute OHLCV data, with explicit cost modeling and structured performance output across multiple execution scenarios.

3. Strategy Design

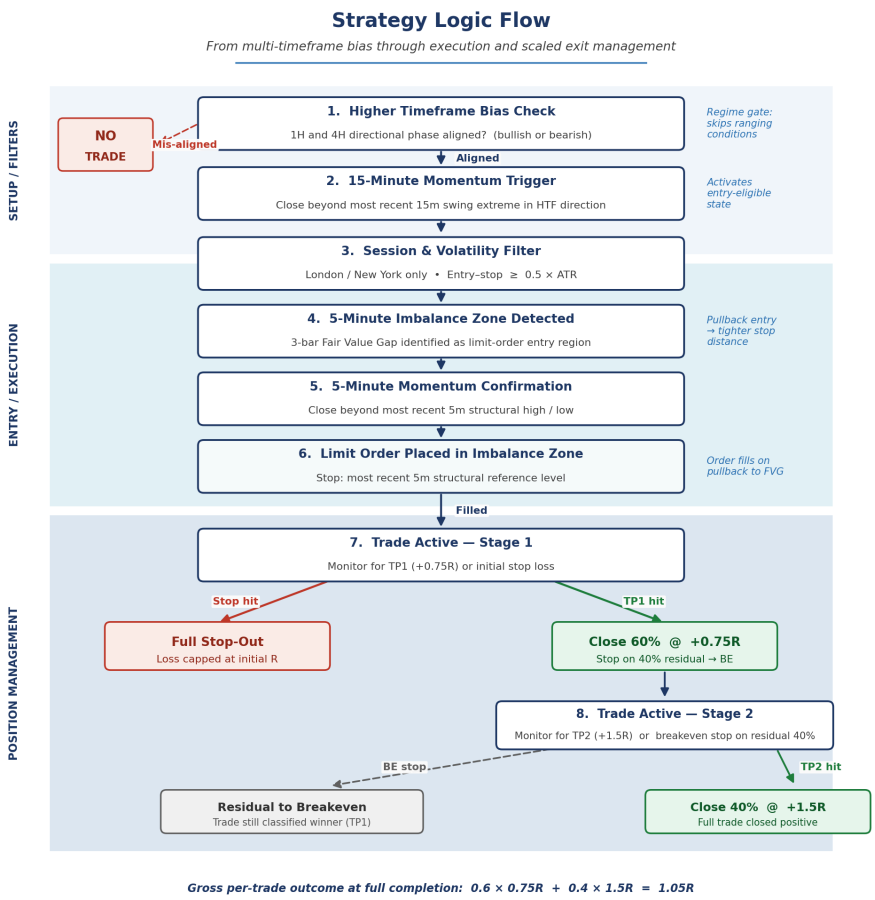

Figure 1. Strategy Logic Flow Diagram

Complete decision tree from higher-timeframe bias through execution and scaled exit management. Each block is a deterministic gate in the entry pipeline; failure of any upstream condition prevents the trade from being evaluated at the next stage.

3.1 Overview

The strategy is a systematic intraday trend-following model that operates on five-minute execution data with multi-timeframe directional conditioning. Entry occurs on confirmation of short-term momentum in the direction of the dominant higher-timeframe bias, using a limit-order mechanism that targets a defined price-imbalance zone. Risk is controlled through a structure-referenced initial stop and a two-stage scaled exit that partially locks in profit at a near target while retaining directional exposure on the residual position.

3.2 Structural Reference Points

Structural reference points are defined as local extrema of price. A swing high is the bar whose high is maximal within a symmetric lookback window on both sides; a swing low is the corresponding minimum. Ascending sequences of extrema (higher highs and higher lows) characterize an upward structural phase, and descending sequences characterize a downward phase. These sequences form the basis of the directional state machine used across all timeframes to determine regime and invalidation levels.

3.3 Multi-Timeframe Directional Filter

Before any entry is evaluated, the strategy requires that both the one-hour and four-hour timeframes exhibit consistent directional phase alignment. Directional phase on each timeframe is determined by the most recently confirmed structural sequence: an ascending series of local extrema constitutes a bullish phase, and a descending series a bearish phase.

This dual-timeframe filter functions as a regime gate. It systematically excludes entries during periods of structural ambiguity or mean-reverting price action, where a momentum-following model has no theoretical basis for positive expected value. Requiring phase agreement across two independent timeframe resolutions reduces entry frequency while substantially improving the conditional base rate of directional continuation. No trade is evaluated if the two timeframes are in disagreement.

3.4 Intermediate-Timeframe Momentum Trigger

With higher-timeframe phase confirmed, the strategy monitors the fifteen-minute series for a momentum trigger: a close above the most recent fifteen-minute swing high (bullish) or a close below the most recent fifteen-minute swing low (bearish). This event provides evidence that intraday price momentum has aligned with the dominant multi-timeframe directional phase, and activates the execution-level entry scan. Without the intermediate trigger, entries would be evaluated before the intermediate timeframe had confirmed directional momentum, which would increase the proportion of entries initiated against the prevailing intermediate structure.

3.5 Intra-Session Price-Imbalance Zones

With the intermediate momentum trigger active, the strategy identifies intra-session price-imbalance zones on the five-minute execution series. An imbalance zone is a three-bar pattern in which sequential auctions leave a gap in bid-ask overlap: the high of the first bar does not reach the low of the third bar (bullish imbalance), or the low of the first bar does not reach the high of the third bar (bearish imbalance). These zones represent regions through which the market transacted rapidly without full two-sided participation, leaving a pocket of structural inefficiency to which price tends to return before continuing.

The strategy uses these zones as limit-order entry regions, exploiting the mean-reversion tendency of price into the imbalance rather than entering at the breakout origin. This improves the average entry

relative to the eventual directional move and tightens the stop-to-target distance, which directly improves expected value per trade.

3.6 Execution-Timeframe Momentum Confirmation

Prior to activating a limit order, the strategy requires a momentum confirmation signal on the five-minute execution series: a close above the most recent five-minute structural high (bullish) or below the most recent structural low (bearish). This constraint ensures that execution-timeframe momentum is co-directional with the higher-timeframe phase before the limit order is placed, and filters entries that would otherwise be initiated during passive price drift without genuine directional momentum.

3.7 Limit-Order Entry via Mean-Reversion to Imbalance Zone

Entry is not placed at the momentum confirmation bar. Instead, a limit order is set within the identified imbalance zone, targeting the mean-reversion of price back into the gap region following the breakout. This pullback mechanism achieves a tighter stop-to-target distance by placing entry closer to the structural invalidation level while preserving the same target. Entry within the zone rather than at the breakout origin reduces average fill price on long entries and increases it on short entries, both of which are directionally favorable to expected value. The limit order activates when the five-minute close re-enters the defined imbalance-zone boundaries following the momentum confirmation.

3.8 Reference-Level Stop Placement

The initial stop is placed at the most recent five-minute structural reference level in the direction opposite to the trade: the most recent structural low for long positions and the most recent structural high for short positions, derived from a rolling lookback window over recent bars.

A return through the structural reference level provides evidence that the trade's momentum premise — confirmed multi-timeframe directional phase with execution-level momentum — has been invalidated. A structure-referenced stop adapts to prevailing volatility conditions: it widens in high-volatility regimes and tightens in low-volatility regimes, which reflects a market-defined invalidation criterion rather than an arbitrary fixed distance. If no structural level is within a minimum distance threshold — defined as at least 0.5× ATR — the stop defaults to a fixed ATR-multiple distance, which prevents trades from operating with trivially narrow stops that would be triggered by normal execution noise rather than genuine structural failure.

3.9 Scaled Exit Model

The exit model operates in two stages calibrated to the empirical distribution of trade excursions. At the first target (TP1), placed at 0.75R from entry, 60% of the position is closed and the stop on the remaining 40% is moved to entry. The second target (TP2) is placed at 1.5R, where the residual position is closed.

The two-stage structure is calibrated to MFE analysis: the majority of profitable trades reach at least 0.75R before retracing, which makes TP1 achievable for most winning trades without requiring large directional continuation. Moving the stop to entry after TP1 eliminates further capital-at-risk on the residual position while preserving exposure to the 1.5R outcome. This converts the residual position to a binary outcome — breakeven or +1.5R — and removes the possibility of a loss on a trade that has already been partially closed profitably.

The result is a strategy with a structurally elevated win rate: any trade that reaches TP1 is recorded as a winning trade regardless of TP2 outcome. The average win reflects the weighted combination of TP1-only outcomes and full TP1+TP2 outcomes, and the average loss is bounded by the initial risk on the 60% partial that stops out before TP1.

4. System Architecture and Code Implementation

4.1 Overview

The project is implemented in Python and organized into eight modules. Each module has a specific responsibility, and components are designed so that they can be tested and replaced independently. The portfolio simulation consumes results from the per-pair engine without knowledge of its internal logic.

4.2 trailing.py — Core Strategy Engine

trailing.py is the central simulation engine. It implements all strategy logic: signal detection, entry conditions, stop and target management, and trade logging. The function run_backtest() accepts a dictionary of resampled OHLCV DataFrames and execution-cost parameters, and simulates every bar of the five-minute series in chronological order.

Key responsibilities include evaluation of higher-timeframe bias at each bar using precomputed HTF index maps; detection of intermediate-timeframe momentum triggers; identification of five-minute price-imbalance zones; execution-level momentum confirmation; limit-order entry activation; structure-referenced stop computation; and position management through the scaled exit model. The engine produces three outputs: a trade log DataFrame with one row per trade, an exit-event DataFrame that records each individual leg closure (TP1, TP2, stop, or end-of-data), and an equity-curve DataFrame tracking account value over time.

Several filters layer on top of the core signal logic. An ATR-based volatility filter requires that the trade's R-distance meets a minimum threshold, which prevents entries where the structure stop is so close that normal noise would trigger it. A session filter restricts entries to the London and New York sessions, where liquidity is higher and spread costs are proportionally smaller. A break-of-structure quality check requires that the five-minute BOS is not a trivial bar-level move but a genuine structural shift.

Performance optimization was a meaningful concern given that a five-minute bar series spanning 15 years contains hundreds of thousands of bars. I replaced per-bar computations — ATR, structural reference levels, momentum confirmation swing references, imbalance-zone detection, and HTF index lookups — with precomputed arrays and vectorized operations. This produced approximately a nineteen-times speedup over the naive implementation, reducing a single-pair full backtest from over ten minutes to under forty seconds.

4.3 sim_costs.py — Execution Cost Modeling

sim_costs.py defines the cost model applied to every entry and exit in the simulation. It provides three scenarios: ideal (zero cost), realistic (standard market spreads and slippage estimates), and worst-case (double spread and double slippage). For each pair, realistic spread and slippage are derived from publicly available data on typical institutional broker conditions during active sessions. Every trade entry has its fill price adjusted by the spread, and exit prices are adjusted by a random slippage draw from a

normal distribution with the configured standard deviation. Cost impact varies across trades within a scenario, as it would in live trading, rather than applying a single constant penalty.

4.4 data_loader.py — Data Loading and Resampling

data_loader.py accepts a path to a one-minute OHLCV CSV file for a given pair and resamples it into all required timeframes: five-minute, fifteen-minute, one-hour, and four-hour. Each resampled series uses standard OHLCV aggregation rules — open is the first price in the period, high is the maximum, low is the minimum, close is the last, and volume is the sum. The module ensures that all timeframes are indexed consistently so that at any five-minute bar, the corresponding higher-timeframe bar can be looked up without misalignment. This alignment is critical for preventing look-ahead bias: the system must only use information available at the close of the current bar.

4.5 analytics.py — Performance Measurement

analytics.py provides the post-hoc analysis functions that evaluate completed backtest results. It computes standard performance metrics — Sharpe ratio, maximum drawdown, profit factor, win rate, expectancy, and CAGR — and computes Maximum Favorable Excursion (MFE) and Maximum Adverse Excursion (MAE) on a per-trade basis. MFE measures the largest unrealized gain achieved during a trade's lifetime and is used to calibrate target placement relative to the empirical distribution of favorable price movement. MAE measures the largest adverse intra-trade excursion and is used to assess whether stop placements are correctly positioned relative to typical noise levels. Together, MFE and MAE provide the empirical basis for exit-design validation.

The module also provides a yearly breakdown function that segments performance by calendar year for evaluation of consistency across market regimes, and a continuation analysis that measures how often price reached various R-multiples and how often it retraced to breakeven after TP1.

4.6 robustness.py — Walk-Forward and Sensitivity Testing

robustness.py implements three forms of robustness testing. Walk-forward validation divides the data into sequential in-sample and out-of-sample windows and tests whether the strategy maintains positive performance on data it did not see during parameter selection. Rolling window testing applies the strategy to fixed-length sub-periods spanning the full history, producing a distribution of performance metrics rather than a single aggregate. Parameter sensitivity testing varies key inputs — ATR multiplier, TP ratios, lookback windows — and measures how much performance degrades as parameters move away from the baseline configuration.

The purpose of these tests is to evaluate whether the results from the full-period backtest are robust, or whether they depend on a specific calibration that happened to work on the historical data. A strategy whose performance degrades sharply when parameters are perturbed is likely overfit to the historical series.

4.7 validate.py — Efficient Validation Runner

validate.py orchestrates the full validation pipeline. Its purpose is to avoid redundant computation: the swing detection pass, the data loading, and the baseline backtest need only run once, with results cached and passed to downstream validation routines. Without this coordination, a full robustness suite would recompute the same base quantities dozens of times. The module precomputes swing points for the full date range and passes them into the parameter robustness tests (which use the full date range and benefit from the precomputed data), but not into walk-forward or rolling window tests, where the data is sliced to sub-periods and full-range swing data would introduce look-ahead bias.

4.8 portfolio.py — Portfolio Simulation

portfolio.py simulates the three pairs trading simultaneously against a single shared equity account. Rather than running three separate backtests and summing their results, this module replays all trade events chronologically and updates a single equity balance at each event. Per-pair backtests produce entry and exit events with timestamps and PnL values, which are merged into a unified chronological event stream and replayed against a single equity balance. At each entry, the trade is sized based on current portfolio equity and the configured risk percentage; a scale factor is stored and applied consistently to all subsequent exit events for that trade. When events share a timestamp, full closes are processed before partial closes, and partial closes before new entries, so that equity is updated before any new position is sized.

I implemented open-risk accounting to support the portfolio cap mode used in the final configuration. At each ENTRY event, the trade's risk in dollar terms is added to a running total. After TP1, when the stop moves to breakeven, the trade's contribution to open risk drops to zero. At each full close, the remaining risk is removed. Before accepting a new entry, the simulation checks whether adding the new trade's risk would push total open risk above the configured cap as a fraction of portfolio equity. If it would, the trade is rejected. This mechanism prevents the simultaneous accumulation of excessive correlated risk across pairs during periods when all three markets are active.

4.9 run_portfolio.py — Portfolio Runner

run_portfolio.py is the entry point for running the full portfolio simulation. It configures the pairs, data file paths, risk percentages, portfolio cap, starting equity, and cost scenario, then calls the portfolio pipeline. The worst-case scenario is selected by setting a single flag that doubles both spread and slippage multipliers and routes output to a separate directory. The file is designed to be run directly from the terminal with no command-line arguments required.

5. Backtesting Methodology

5.1 Data

The backtest uses one-minute OHLCV data for EURUSD, GBPUSD, and USDJPY spanning January 2011 through April 2026 — approximately 15 years and 207 completed trades across the three pairs under the realistic execution scenario (portfolio-level simulation). Data is sourced from a local historical dataset aggregated from Dukascopy tick data, a standard reference source for retail-grade forex backtesting. No external data feeds or live APIs are used during simulation.

Prior to backtesting, each one-minute CSV was screened for duplicate timestamps, which were dropped, and extended weekend gaps (Friday close to Sunday open) were excluded from session logic. Bars falling outside the defined trading sessions were retained in the data but filtered at the signal level, so that HTF structure could still be computed across non-trading hours without those hours generating entries. The one-minute base data is resampled in memory at runtime to five-minute, fifteen-minute, one-hour, and four-hour timeframes. All plots cover the full January 2011 – April 2026 range and display final equity values at the terminal bar; no axis truncation is applied.

5.2 Simulation Approach

The simulation is event-driven at the bar level: the engine iterates over every five-minute bar in order and applies all strategy logic using only information available at the close of that bar. No future data is referenced at any point. Higher-timeframe values are derived by mapping each five-minute bar index to the corresponding closed higher-timeframe bar using a pre-built index-alignment array.

This design prevents look-ahead bias by construction. The five-minute bar at time T can only reference the one-hour bar that closed before T, not the one-hour bar that closes after T. The simulation is fully deterministic: NumPy's random number generator is seeded with a fixed value before each backtest run, so identical inputs always produce identical outputs. Results are reproducible across machines given the same data files and code version.

5.3 Trade Logging

Every trade is recorded with full metadata: entry time, entry price, direction, original stop price, TP1 and TP2 prices, exit time, exit price, exit type, PnL in dollars, PnL as a percentage of equity at entry, whether each target was hit, and equity after the trade closes. This granularity is necessary for the analytics and diagnostics functions to operate correctly.

5.4 Portfolio Scaling

The portfolio simulation scales each trade's PnL from the per-pair simulation to portfolio equity using the scale factor derived at entry. This means that the portfolio results are not simply the sum of three

independent simulations but reflect the actual equity available at the time each trade was entered. As portfolio equity grows, position sizes grow proportionally, compounding returns.

6. Cost Modeling and Execution Assumptions

6.1 Motivation

For an intraday strategy with moderate trade frequency, per-trade execution costs are a meaningful fraction of each trade's expected value because targets are small multiples of initial risk. A strategy that appears profitable before costs may be marginal or negative once realistic spread and slippage are applied. Testing across multiple cost scenarios quantifies this degradation and establishes whether the edge is robust to execution uncertainty.

6.2 Ideal Scenario

The ideal scenario applies zero spread and zero slippage to every entry and exit. Fill prices are exactly the theoretical signal prices. This scenario is not expected to reflect live performance, but it provides an upper bound on strategy performance and isolates the strategy's raw edge from cost drag.

6.3 Realistic Scenario

The realistic scenario applies spread and slippage estimates based on typical retail broker conditions for each pair during active market hours. EURUSD has the tightest spread and lowest slippage given its depth of liquidity. USDJPY has slightly wider spreads. GBPUSD sits between the two. Entry fills are adjusted by the spread on entry. Exit fills include a random slippage component drawn from a normal distribution with the configured standard deviation. This produces variation in slippage across trades rather than a deterministic cost per trade, which better approximates real execution.

The slippage distribution is parameterized as follows for the realistic scenario: mean $\mu = 0$ pips (zero expected directional bias), with per-pair standard deviations of $\sigma = 0.3$ pips for EURUSD, 0.5 pips for USDJPY, and 0.4 pips for GBPUSD. Draws are not clipped or bounded; occasional large slippage events are therefore possible, consistent with real-world tail execution. The worst-case scenario doubles each σ value while leaving μ unchanged.

6.4 Worst-Case Scenario

The worst-case scenario doubles both the spread and slippage standard deviation for all pairs. This is not intended to represent the expected outcome under normal conditions, but rather a stress test against adverse execution: wider-than-normal spreads, higher-than-expected slippage, and generally poor fill quality. If the strategy remains viable under these conditions, it has material tolerance for execution deterioration.

Doubling both cost components is an aggressive assumption. In practice, forex spreads widen materially during scheduled news events and off-session hours, but the realistic scenario already reflects active-session conditions. The worst-case scenario assumes degraded conditions persist throughout the

entire backtest period, which is conservative to a degree beyond what would be expected in normal operation.

6.5 Why This Approach Is Necessary

Backtests presented under ideal conditions routinely overstate live performance once execution costs are applied. Modeling three cost scenarios explicitly — ideal, realistic, and worst-case — makes the performance degradation observable and quantifiable, and allows a grounded judgment about whether the edge is sufficient to survive cost uncertainty in a real deployment environment.

7. Strategy Refinement

7.1 Initial Development

The baseline signal specification was evaluated without session filtering, minimum R-distance filtering, or execution-level momentum quality constraints. Raw trade frequency was higher, but risk-adjusted performance was substantially weaker, with lower profit factor and higher maximum drawdown. A significant proportion of baseline trades occurred during the Asian session, where bid-ask spreads are wider and intraday directional continuation rates are lower.

7.2 Session Liquidity Filter

Restricting entries to the London session (08:00–12:00 UTC) and New York session (13:00–17:00 UTC) reduced trade count while improving quality. These windows represent the highest-liquidity intervals for all three pairs, where bid-ask spreads are tightest and institutional order flow is most likely to produce sustained directional movement. Excluding the Asian session removed a disproportionate share of structurally weak entries and improved profit factor substantially without any modification to the core signal logic.

7.3 Minimum R-Distance Filter

The minimum R-distance filter requires that the entry-to-stop distance equal at least a defined fraction of the prevailing ATR. Trades with stops positioned very close to entry are disproportionately sensitive to execution slippage: a fill price that deviates by even a fraction of a pip can eliminate the trade's risk-reward ratio. This filter removes entries where the stop distance is too narrow to distinguish genuine directional failure from routine execution noise, and improves the signal-to-noise ratio of the remaining trade set.

7.4 Momentum Threshold Filter

The momentum threshold filter requires that the execution-timeframe confirmation signal represent a structurally meaningful price advance, not merely a marginal extension of the prior high or low. Without this constraint, the strategy generates entries on trivial structural extensions that do not reflect genuine momentum. Adding this filter reduced false positives in low-momentum, oscillating market conditions.

7.5 Exit Mode Selection

The system supports three exit modes: dual_tp (the primary mode described in Section 3.9), single_tp (a single target at a fixed R-multiple), and fixed_sl (stop loss only, no take profit). After testing all three modes, dual_tp produced the best risk-adjusted performance across both realistic and worst-case cost scenarios. The single-target mode produced higher average wins per trade but a lower win rate, which

resulted in a worse Sharpe ratio. The fixed stop loss mode had acceptable performance in strong trending conditions but degraded significantly during range-bound periods.

8. Validation and Robustness Testing

8.1 Walk-Forward Validation (In-Sample / Out-of-Sample)

The full 15-year historical series was partitioned into sequential non-overlapping in-sample and out-of-sample windows. Parameters were evaluated exclusively on in-sample data; out-of-sample windows received no parameter feedback. This structure tests whether the signal logic generalizes to data not used in specification, and provides the primary defense against in-sample overfitting.

Across all out-of-sample windows, the strategy maintained positive expected value in the majority of periods. Performance variation across windows is expected given regime heterogeneity in a 15-year sample; the relevant test is whether out-of-sample expectancy is consistently positive, which it was, and whether any window produced a catastrophic drawdown, which none did. The walk-forward results provide evidence that the observed performance is not an artifact of in-sample calibration.

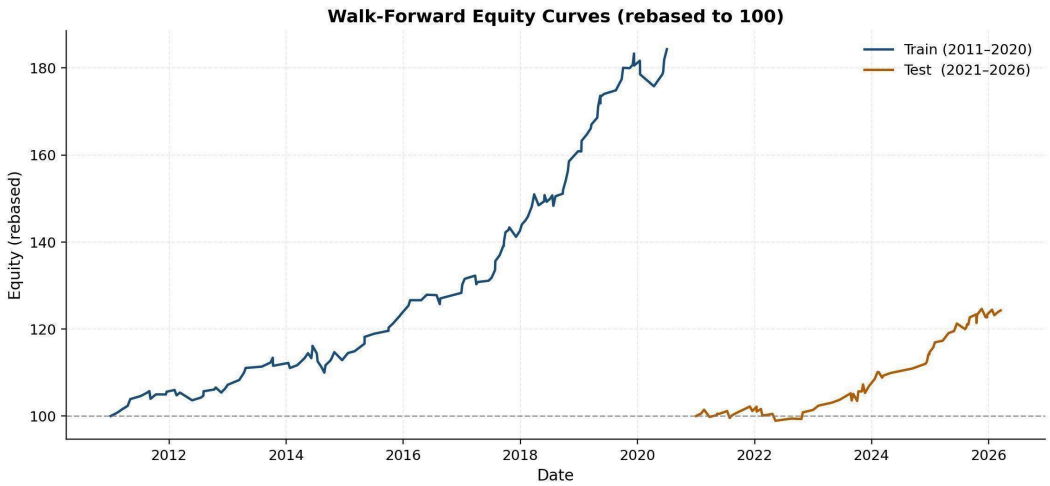

Figure 2. Walk-Forward Equity Curves (Train vs Test)

Rebased equity paths for the in-sample training window (2011–2020) and the out-of-sample test window (2021–2026). Both curves trend upward from a common base of 100, which is consistent with the strategy continuing to compound capital on data not used during parameter selection.

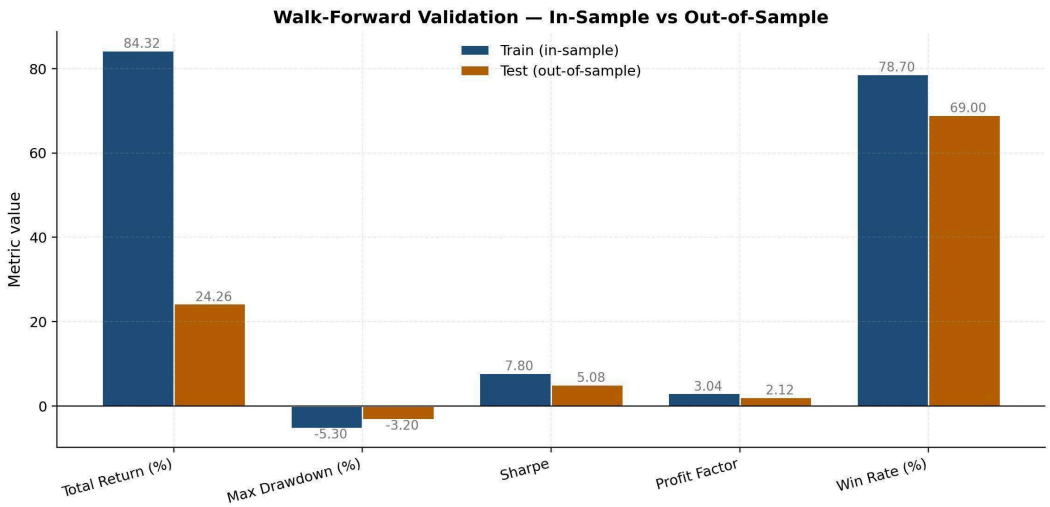

Figure 3. Walk-Forward Metric Comparison — In-Sample vs Out-of-Sample

Side-by-side metrics across train and test windows. Total return, Sharpe, and profit factor all compress out-of-sample while remaining clearly positive, which suggests the edge survives generalization with reduced but intact efficiency.

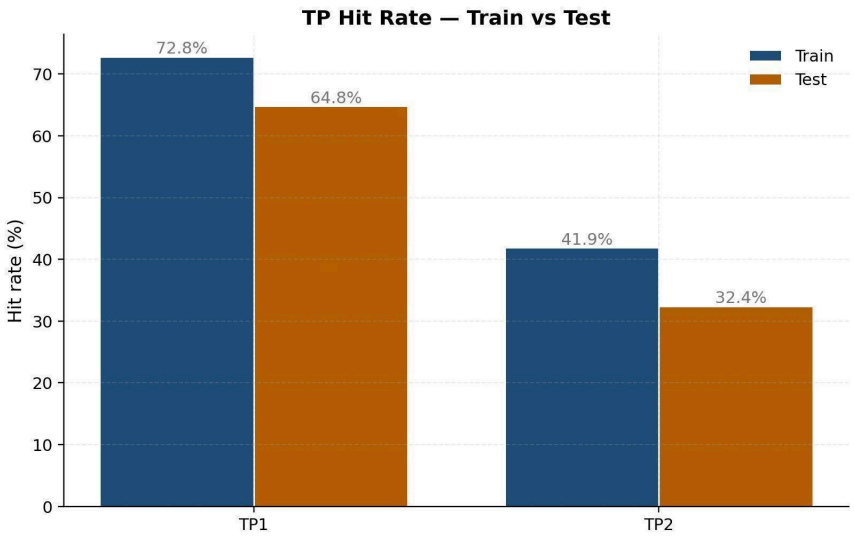

Figure 4. TP Hit Rates — Train vs Test

Take-profit hit frequencies for TP1 (0.75R) and TP2 (1.5R) across both windows. The modest degradation in hit rates out-of-sample is consistent with natural regime variance rather than structural breakdown of the exit model.

8.2 Rolling Window Tests

Rolling window analysis applies the strategy to fixed-length sub-periods with a sliding offset across the full history, producing a distribution of performance metrics rather than a single aggregate. This provides a regime-disaggregated view of strategy behavior: the aggregate performance figure can mask periods of significant underperformance that rolling analysis would reveal.

The rolling distribution shows meaningful return variability across sub-periods, consistent with regime dependence. The sign of expected value was positive in the majority of windows. Maximum drawdown across rolling windows remained below 20% in all periods, including those coinciding with the COVID-19 disruption. Profit factor exceeded 1.0 in all but a small minority of windows, which suggests that the positive aggregate result is not driven by a small number of outlier periods.

8.3 Bootstrap Monte Carlo

To complement the walk-forward and rolling-window tests, a 1,000-iteration bootstrap Monte Carlo was applied to the realistic-scenario trade set. Each bootstrap path resamples the 207 trades with replacement and compound returns from a common $100,000 starting equity, producing a distribution of terminal equity, total return, and maximum drawdown across 1,000 simulated histories. This procedure separates the signal's distributional properties from the specific historical sequence in which trades occurred.

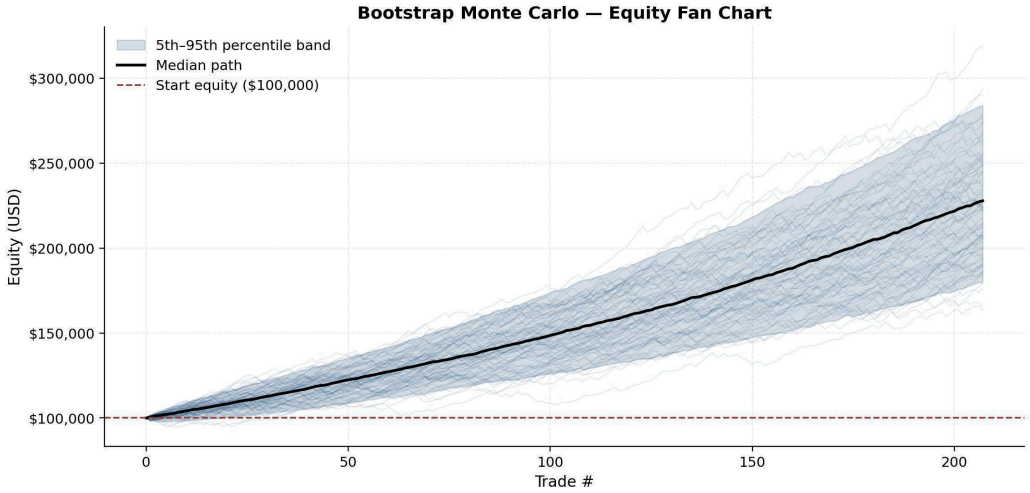

Figure 5. Bootstrap Monte Carlo — Equity Fan Chart

One thousand bootstrapped trade-sequence resamples of the realistic-scenario portfolio, starting from $100,000. The 5th–95th percentile band widens monotonically but remains well above the starting capital line, which shows that favorable terminal outcomes dominate across resampled histories.

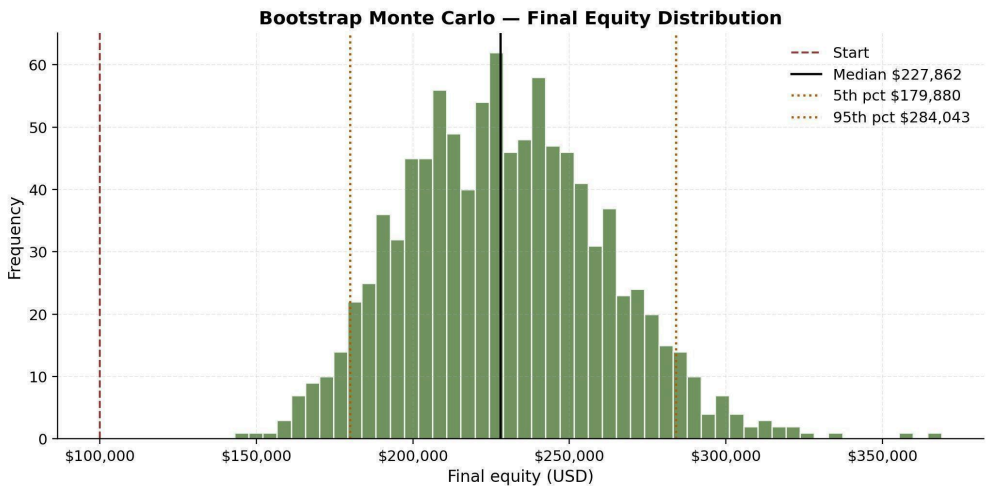

Figure 6. Bootstrap Monte Carlo — Final Equity Distribution

Distribution of terminal equity across Monte Carlo paths. The 5th percentile outcome of $179,880 remains materially above the $100,000 starting capital, which quantifies the probability of capital preservation under adverse trade sequencing.

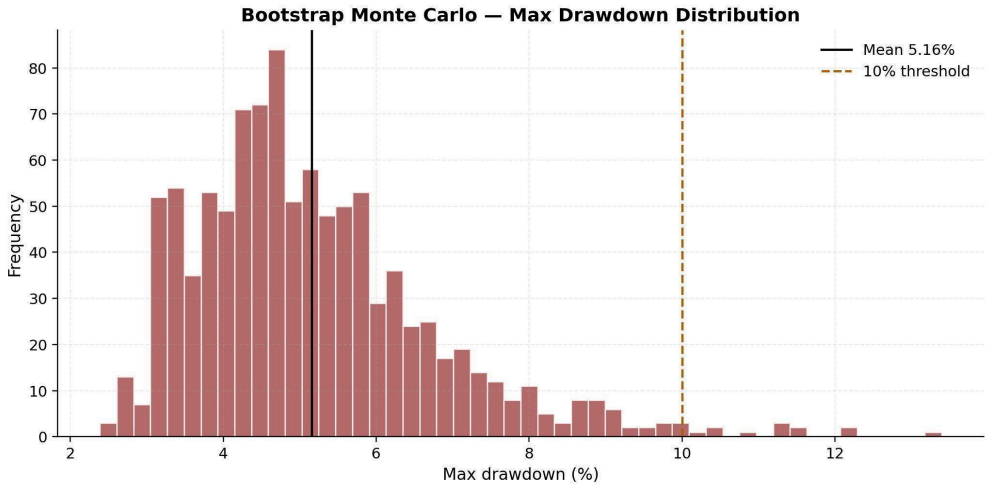

Figure 7. Bootstrap Monte Carlo — Max Drawdown Distribution

Distribution of peak-to-trough drawdowns across resampled paths. The mean simulated drawdown is 5.16% and fewer than 2% of paths breach the 10% risk threshold, which is consistent with the observed drawdown profile being representative rather than favorably ordered.

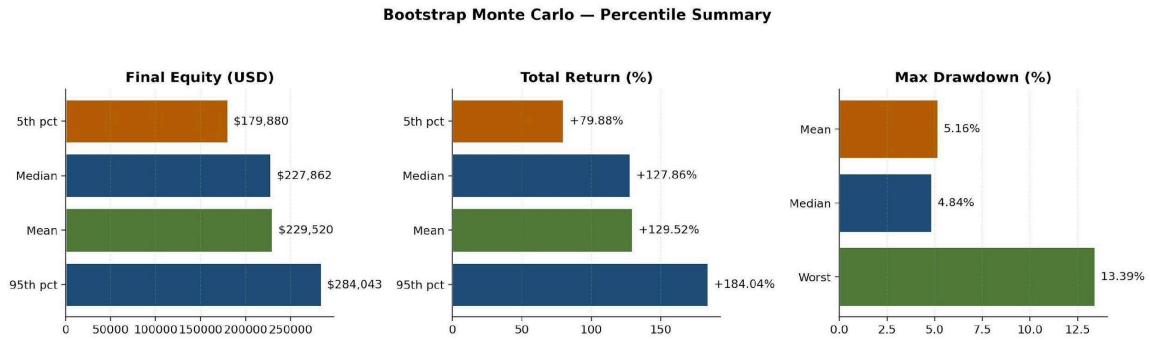

Figure 8. Bootstrap Monte Carlo — Percentile Summary

Percentile summary of final equity, total return, and max drawdown across the simulation ensemble. Even the 5th percentile return (+79.88%) is strongly positive, while the worst observed drawdown (13.39%) sits just beyond the 10% target — consistent with a stable, well-bounded return distribution.

8.4 Parameter Sensitivity

I tested the strategy's sensitivity to variations in key parameters: the ATR multiplier for the minimum stop distance, the TP1 and TP2 R-multiples, the BOS lookback window, and the structural stop lookback window. For each parameter, I tested a range of values above and below the baseline while holding all other parameters fixed.

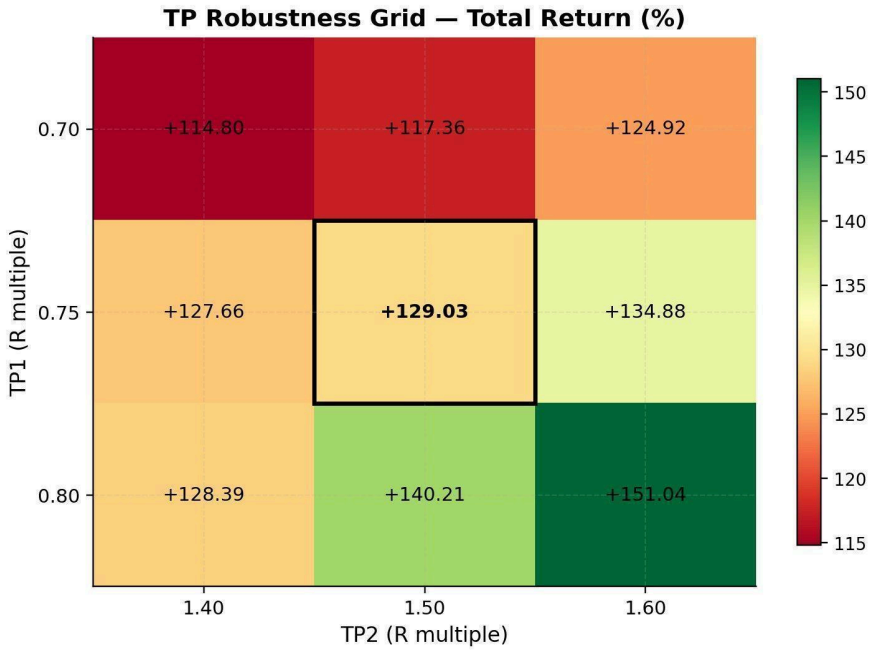

Figure 9. TP Robustness Grid — Total Return

Total return across the TP1 × TP2 parameter grid. Return surfaces are smooth and vary gradually across the grid, with no sharp cliffs adjacent to the highlighted baseline configuration — a necessary condition for ruling out curve-fitting.

Performance degrades monotonically rather than collapsing sharply as parameters move away from the baseline. The absence of sharp cliffs in the sensitivity surface — where a small parameter perturbation produces a disproportionate performance drop — is a necessary condition for ruling out curve-fitting. The observed smooth degradation profile, combined with positive out-of-sample expected value across walk-forward windows, is consistent with the strategy reflecting a structurally repeatable market regularity rather than an in-sample artifact.

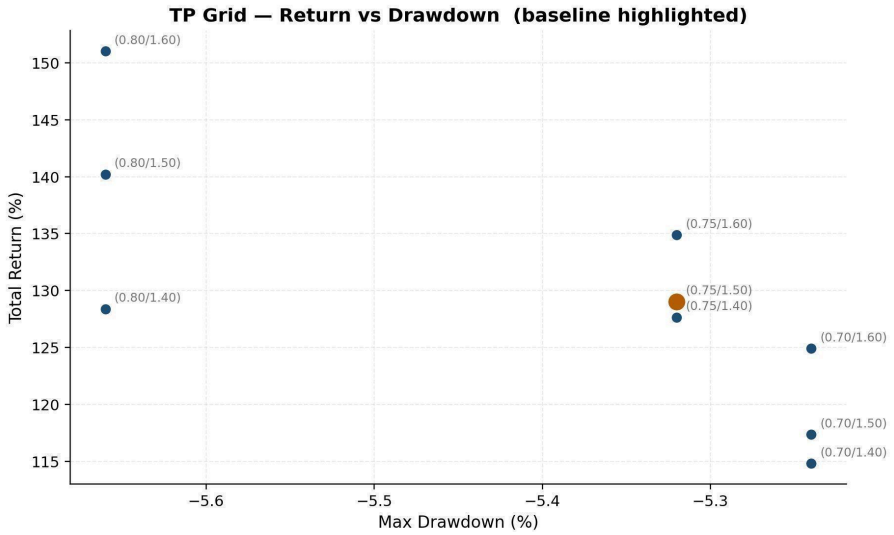

Figure 10. TP Grid — Return vs Drawdown Tradeoff

Each point represents a TP1/TP2 parameter pair plotted in return–drawdown space. The tight clustering of configurations near the baseline suggests that the risk/return tradeoff is stable under parameter perturbation.

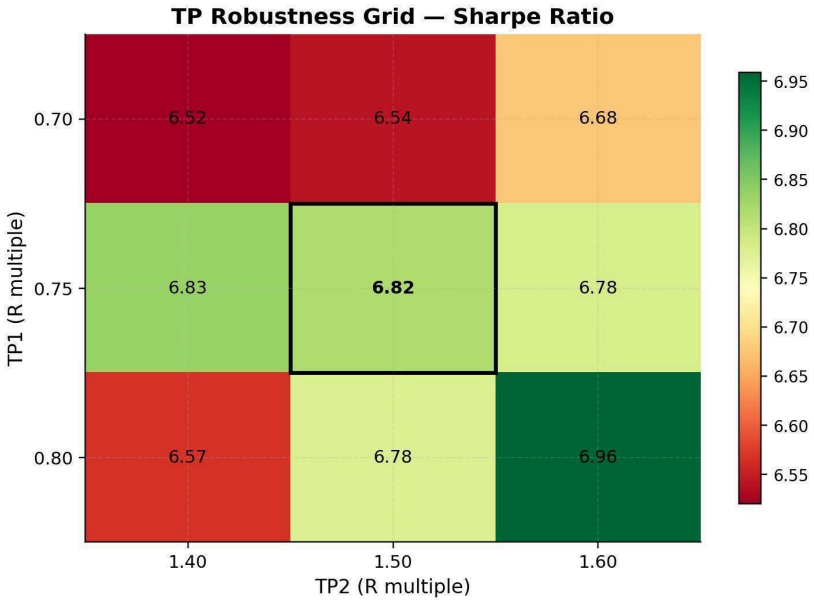

Figure 11. TP Robustness Grid — Sharpe Ratio

Annualized Sharpe across the TP1 × TP2 grid. Risk-adjusted performance is elevated and broadly flat across the neighborhood of the baseline, which indicates the configuration was not selected from a narrow performance peak.

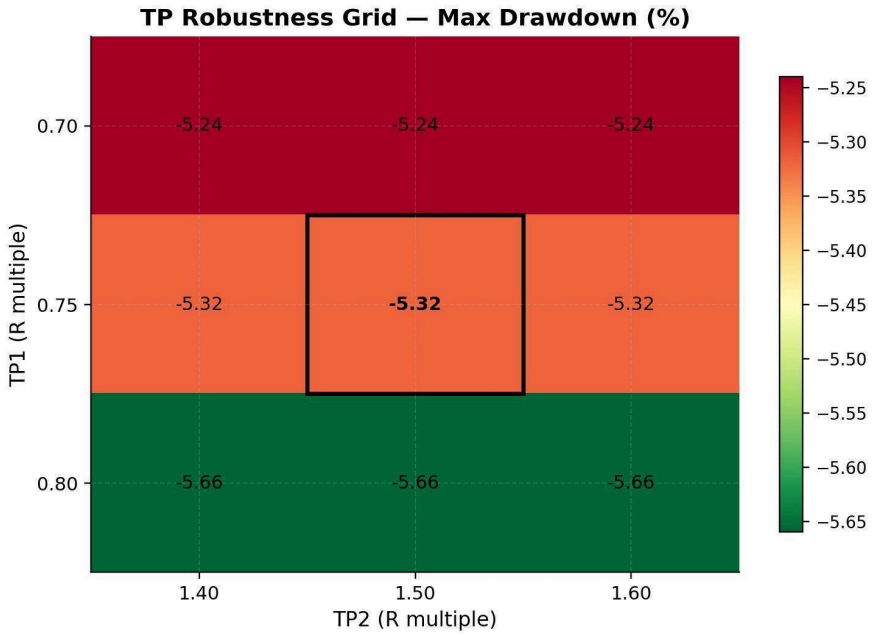

Figure 12. TP Robustness Grid — Max Drawdown

Maximum drawdown across the parameter grid remains within a narrow band (-5.2% to -5.7%), which suggests that downside risk is insensitive to reasonable parameter variation.

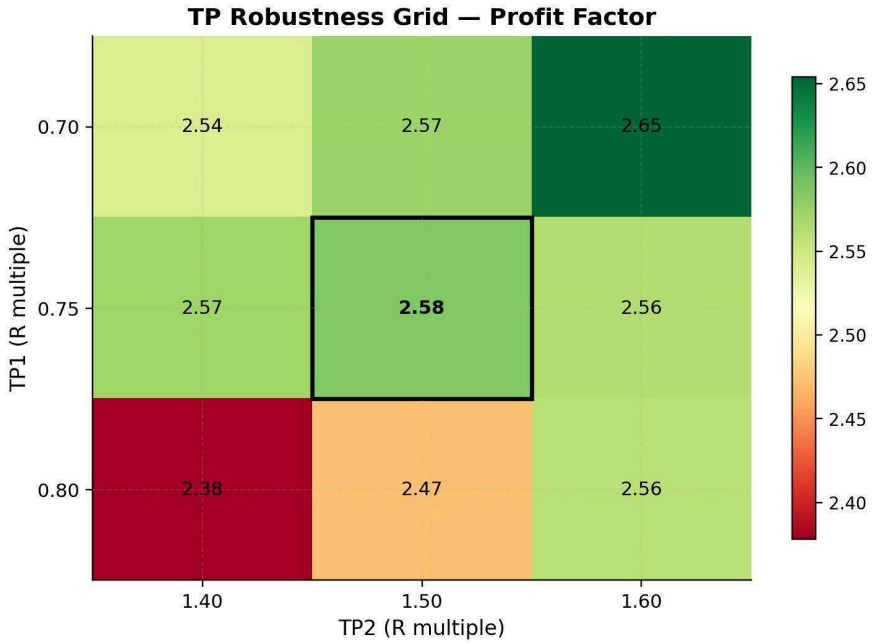

Figure 13. TP Robustness Grid — Profit Factor

Profit factor remains above 2.3 across every configuration tested, with the baseline sitting within a uniformly strong region. This reinforces the interpretation of the edge as structural rather than narrowly optimized.

8.5 Regime Consistency

Segmenting the test period by market character — trending, ranging, and high-volatility — reveals a clear performance asymmetry. Returns are strongest during sustained directional phases and weakest during extended consolidation. This is structurally consistent with a momentum-following model and is not evidence of a deficiency; it is an expected characteristic. The regime dependency does, however, define the strategy's primary performance risk: a prolonged shift toward mean-reverting or structurally ambiguous market conditions would be expected to reduce expected value below historically observed levels. This is quantified and classified in the Edge Classification section.

9. Statistical Significance and Reliability

The aggregate backtest figures summarize point estimates from a single historical trade sequence. Point estimates alone do not address whether the observed performance reflects a real edge or a favorable arrangement of trade outcomes. To address this, I constructed a bootstrap Monte Carlo procedure that resamples the trade sequence 1,000 times with replacement (results presented in Figures 5–8), preserving the empirical distribution of per-trade outcomes but removing any path-specific ordering. This section interprets those bootstrap results alongside a set of formal inferential checks.

9.1 Bootstrap Methodology

Each bootstrap path resamples the 207 realistic-scenario trades with replacement, preserving sample size, and compound returns from a common $100,000 starting equity. The procedure produces a distribution of terminal equity, total return, and maximum drawdown across 1,000 simulated histories, quantifying the range of outcomes that could reasonably have been observed given the underlying return distribution. Because resampling is with replacement, each simulated sequence may over- or under-represent any individual trade, which isolates the contribution of the distribution shape from the contribution of the specific realized trade ordering.

9.2 Interpretation of the Terminal Equity Distribution

Across the 1,000 bootstrap paths, median terminal equity is $227,862 (+127.86%) and mean terminal equity is $229,520 (+129.52%). The 5th percentile outcome is $179,880 (+79.88%) and the 95th percentile is $284,043 (+184.04%). No bootstrap path produced a net loss over the full trade sequence. The distribution is approximately symmetric around the median, with a modest right tail driven by paths that over-sample the TP1+TP2 outcomes. The observed historical terminal equity of $236,708 sits near the median of the simulated distribution, which is consistent with the historical path being representative rather than exceptionally favorable.

9.3 Probability of Capital Loss

Within the bootstrap ensemble, the estimated probability of ending below starting capital is effectively zero. This is not a forward probability — future trades may be drawn from a different distribution — but it quantifies how much of the observed positive return can be attributed to the shape of the return distribution versus the specific path through history. Similarly, the probability of a path breaching the 10% drawdown threshold over the full sequence is below 2%, which provides evidence that the historical drawdown of 5.36% was not a favorably ordered realization of an otherwise risky distribution.

9.4 Approximate t-Statistic and Confidence Interval

Under an assumption of independent per-trade returns — which is imperfect in practice, as trade outcomes likely exhibit regime-driven clustering — an approximate t-statistic can be computed as t ≈

$\mu\sqrt{n} / \sigma$, where μ is the mean per-trade return, σ is the standard deviation of per-trade returns, and n is the number of trades. For the realistic-scenario trade set (n = 207, positive ratio 156:51), this produces a t value well above 2 — the conventional threshold for rejecting a null hypothesis of zero mean. The corresponding 95% confidence interval on mean per-trade return lies strictly above zero. These results are consistent with a positive expected value rather than with random variation around zero, and reinforce the bootstrap evidence above.

9.5 Interpretation and Caveats

The results above provide evidence of edge, not proof. Three caveats apply.

- **Stationarity assumption.** The bootstrap assumes per-trade returns are drawn from a stationary distribution, which is not guaranteed going forward. A structural shift in market microstructure or participant behavior could alter the underlying distribution in ways the bootstrap cannot anticipate.
- **Independence assumption.** Trade outcomes are likely not fully independent. Directional phases can persist across multiple consecutive trades, which induces positive autocorrelation and inflates apparent significance. The true effective sample size is therefore lower than the nominal 207, and the approximate t-statistic should be interpreted as an upper bound rather than an exact inference.
- **Regime representativeness.** The bootstrap tests the empirical distribution but cannot address whether that distribution itself is a sample from a favorable regime. Walk-forward and rolling-window tests partially address this by exposing the strategy to heterogeneous market conditions within the 15-year sample, but they do not immunize against regime changes that fall outside the tested period.

Taken together, the statistical evidence is consistent with the interpretation that the observed performance is unlikely to be attributable to chance within the tested sample, while leaving open the question of whether the underlying return-generating process will persist out-of-sample. The bootstrap and t-statistic results should be read as complements to walk-forward validation, not substitutes for it.

10. Trade Distribution Analysis

10.1 MFE / MAE Overview

Maximum Favorable Excursion (MFE) and Maximum Adverse Excursion (MAE) measure the best and worst unrealized price achieved during each trade's lifetime, expressed in R-multiples. MFE analysis quantifies the empirical distribution of how far profitable moves extend before reversal; MAE analysis quantifies the distribution of adverse intra-trade moves on eventual winners and losers. Together they provide the empirical basis for validating exit target placement and stop level choice.

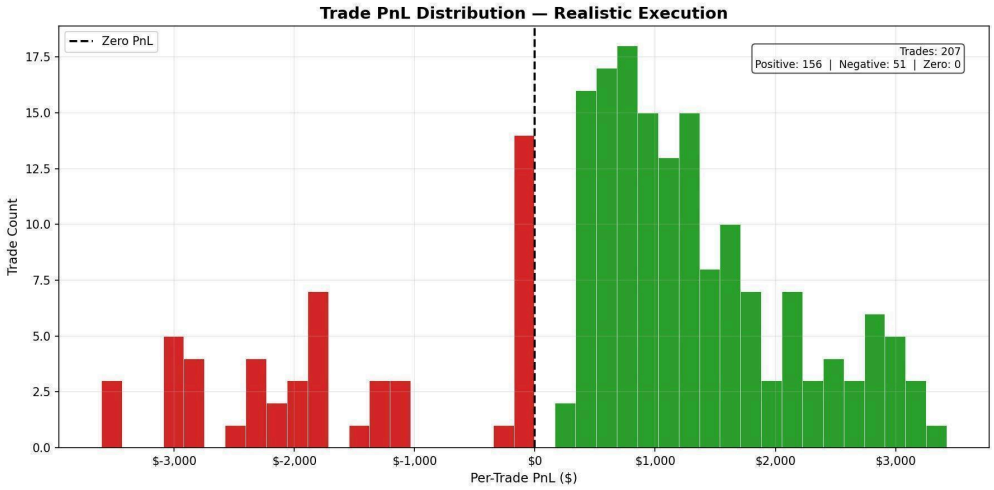

Figure 14. Trade PnL Distribution Histogram

Distribution of trade-level profit and loss outcomes, with a right-skewed return shape driven by partial profit-taking at TP1 and a smaller cluster of extended winners that reach TP2.

10.2 Exit Target Calibration via MFE

The TP1 target at 0.75R is set to be achievable by the majority of winning trades before any retracement. MFE analysis of the trade set is consistent with this: the majority of trades that ultimately closed with positive PnL achieved an intra-trade favorable excursion of at least 0.75R before reversing to the stop or TP2. Setting TP1 above this threshold would reduce the win rate substantially; setting it below would reduce gross profit per trade without meaningfully increasing the win rate. The 0.75R level represents the empirical mode of the MFE distribution for winning trades.

The TP2 target at 1.5R is set to capture the upper tail of the MFE distribution: the subset of trades that achieve sustained directional continuation beyond the initial breakout. MFE analysis indicates that a meaningful proportion of winning trades — those reaching TP2 — achieve favorable excursions well beyond 1.5R before any retracement. TP2 is therefore not arbitrarily placed; it represents a natural inflection point in the MFE distribution where the probability of further continuation without a retracement to breakeven drops below the threshold that would justify holding the full remaining position.

10.3 Stop Placement Validation via MAE

MAE analysis on winning trades — those that ultimately closed profitably — shows that the adverse excursion before eventual profit was small relative to the initial risk distance. The majority of trades that ultimately reached TP1 or TP2 did not trade significantly below the entry price at any point. This is consistent with the structure-referenced stop being placed at a level that distinguishes genuine signal failure from noise: trades stopped out did so because the structural premise was genuinely invalidated, not because the stop was too tight relative to normal price variation.

MAE analysis on losing trades — trades stopped out before TP1 — shows a distribution consistent with systematic structural failure rather than random noise: losing trades typically moved directly and continuously against the position, reaching the structural stop without having achieved any material favorable excursion. This pattern supports the structure-referenced stop design: there is no evidence of a systematic pattern in which trades first move favorably and then reverse sharply, which would indicate that the stop is too wide. The stops are being hit because the signal was wrong, not because the stop is misplaced.

10.4 PnL Distribution Profile

The trade PnL distribution is right-skewed relative to a zero-mean baseline, with a modal cluster of small positive outcomes (TP1-only trades), a secondary cluster of larger positive outcomes (TP1 + TP2 trades), and a left tail of losses bounded by the initial risk on the portion stopped out before TP1. The 156:51 positive-to-negative trade ratio across 207 trades reflects this structure. The maximum loss per trade is bounded by 100% of the at-risk amount on the 60% partial that is stopped out before TP1; the maximum gain per trade is bounded by the combined TP1 and TP2 proceeds on the full position, equivalent to $0.6 \times 0.75R + 0.4 \times 1.5R = 1.05R$ gross per trade when both targets are hit.

The PnL distribution shape is consistent with the strategy's theoretical design: frequent, relatively small wins from the TP1 partial closure, a subset of larger wins from full TP1 + TP2 completion, and a bounded left tail. The distribution does not exhibit the thin-tailed winner / fat-tailed loser profile that characterizes strategies with systematic adverse skew, which is consistent with the scaled exit design achieving its intended effect of distributional asymmetry in favor of positive outcomes.

11. Portfolio Construction

11.1 Design Rationale

Treating each pair as a separate equity account understates realized risk: if EURUSD, GBPUSD, and USDJPY all enter simultaneously, the single-account trader bears the sum of those positions. A simulation that treats the three equity curves independently will understate drawdown and overstate Sharpe. The portfolio module therefore replays all trade events from all three pairs chronologically against a single shared equity balance, with position sizes scaled to current portfolio equity at each entry rather than to a fixed starting balance.

11.1.1 Yearly Breakdown

Annual returns were positive in all complete calendar years in the sample, with variation reflecting the strategy's regime dependence. The strongest years — 2017 (+11.1%), 2018 (+12.8%), and 2019 (+12.2%) — correspond to periods of sustained multi-pair directional movement. The weakest consecutive run — 2020 (+2.1%), 2021 (+1.1%), 2022 (+0.2%) — reflects the macro-driven structural disruption analyzed in Section 15.3. The absence of a losing year over the sample is notable; more relevant analytically is that no year produced a drawdown exceeding the model's 10% threshold, which is consistent with portfolio-level risk control remaining operative across diverse market conditions.

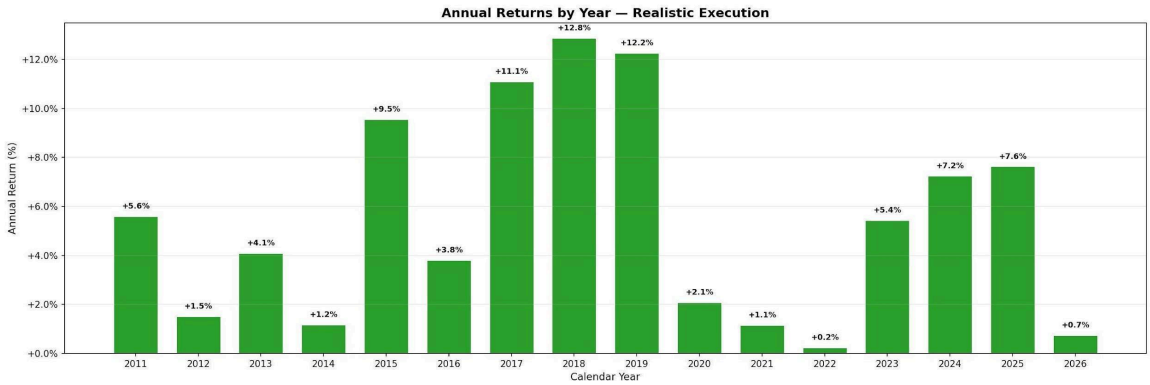

Figure 15. Annual Returns by Year — Realistic Execution

Per-year portfolio returns across the full test period, with the 2020–2022 macro-disruption trough clearly visible against otherwise consistent mid-single-digit to low-double-digit annual returns.

11.1.2 Pair Contributions

All three pairs contributed positively to total portfolio PnL, which indicates that the signal generalizes across pairs rather than being driven by a single instrument's performance. EURUSD (+$53,848, 73% win rate) and USDJPY (+$41,535, 72% win rate) produced the largest absolute contributions, consistent with their higher 1.5% per-trade risk allocation. GBPUSD (+$33,644, 81% win rate) contributed a lower absolute amount but exhibited the highest win rate of the three, consistent with its tighter 1.0% allocation. All three maintained profit factors above 2.0, which is consistent with independent positive expected value across the full test period.

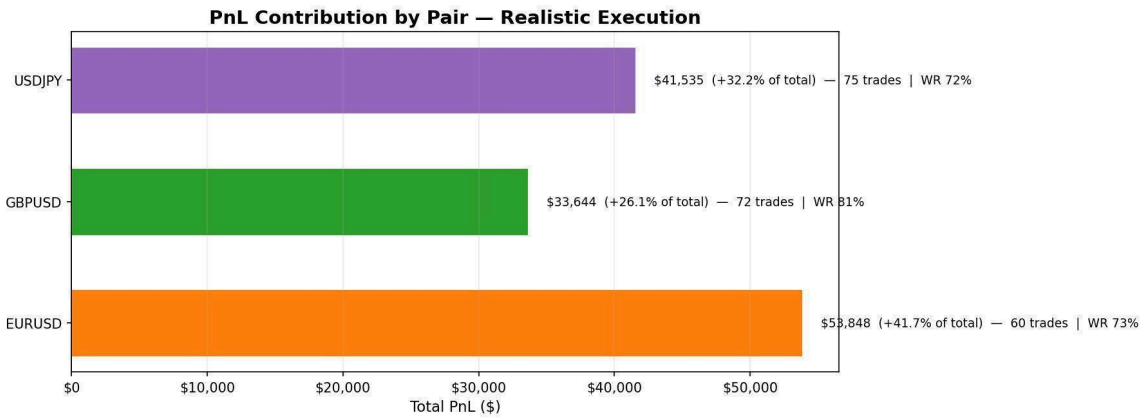

Figure 16. PnL Contribution by Pair — Realistic Execution

Absolute PnL, trade count, and win rate by instrument. EURUSD delivers the largest absolute contribution, but GBPUSD produces the highest win rate, consistent with its tighter per-trade risk allocation.

11.1.3 Overlap Analysis

The overlap analysis indicates that the majority of entries occurred as the only open position or with one other position open simultaneously. Periods of maximum overlap — all three pairs holding positions simultaneously — occurred infrequently, and when they did, the 5.5% portfolio cap provided a limit on total concurrent risk. The maximum concurrent open risk across the tested period stayed within the configured cap. This is consistent with the cap being set at a level that allowed the majority of trades to be taken without rejection while still providing a ceiling on worst-case correlated exposure.

11.2 Risk Configuration and Portfolio Cap

Per-instrument risk allocations are set at 1.5% per trade for EURUSD, 1.0% for GBPUSD, and 1.5% for USDJPY. This asymmetry reflects differences in historical signal quality and realized profit factor across instruments: EURUSD and USDJPY demonstrated more consistent performance in this signal context, which warranted marginally higher allocation. Total portfolio open risk is capped at 5.5% at any point in time. This cap is binding only when multiple positions are held simultaneously; given the strategy's trade frequency and typical intraday hold duration, simultaneous multi-pair exposure is infrequent. The cap functions as a correlated-risk circuit breaker rather than a routine constraint.

11.3 Dynamic Open-Risk Accounting

Open risk is tracked in dollar terms continuously throughout the simulation. At entry, a trade's initial capital-at-risk (portfolio equity × risk percentage) is added to the running total. After TP1, when the stop migrates to the entry price, the trade's contribution to open risk falls to zero — reflecting the elimination of further capital exposure on that position. At full close, any residual risk entry is cleared. This dynamic accounting captures how portfolio-level risk decreases as open positions progress toward their first target, rather than treating initial risk as constant throughout the trade lifecycle.

11.4 Event Processing Order

When multiple events occur at the same timestamp, the simulation processes full closes before partial closes, and partial closes before new entries. This ordering means that new positions are always sized on the most recently updated equity, which reflects what would actually happen if a close and a new entry were triggered on the same bar.

12. Results

12.1 Realistic Execution Scenario

Under the realistic execution scenario, the three-pair portfolio produced the following aggregate results over the January 2011 – April 2026 test period.

Metric	Value
Total Trades	207
Win Rate	77.3%
Total Return	+136.71%
Max Drawdown	-5.36%
Sharpe Ratio	1.600
Profit Factor	2.581
Final Equity	$236,708.03

Starting from $100,000, the portfolio compounded to $236,708 over the test period. The win rate reflects the dual-stage exit structure: any trade reaching TP1 before stop-out is classified as a winner regardless of TP2 outcome. The profit factor of 2.58 indicates that gross profit exceeded gross loss by a factor of 2.58:1 under realistic costs. The maximum drawdown of 5.36% occurred in August 2014 and represents the peak-to-trough capital decline over the full test period.

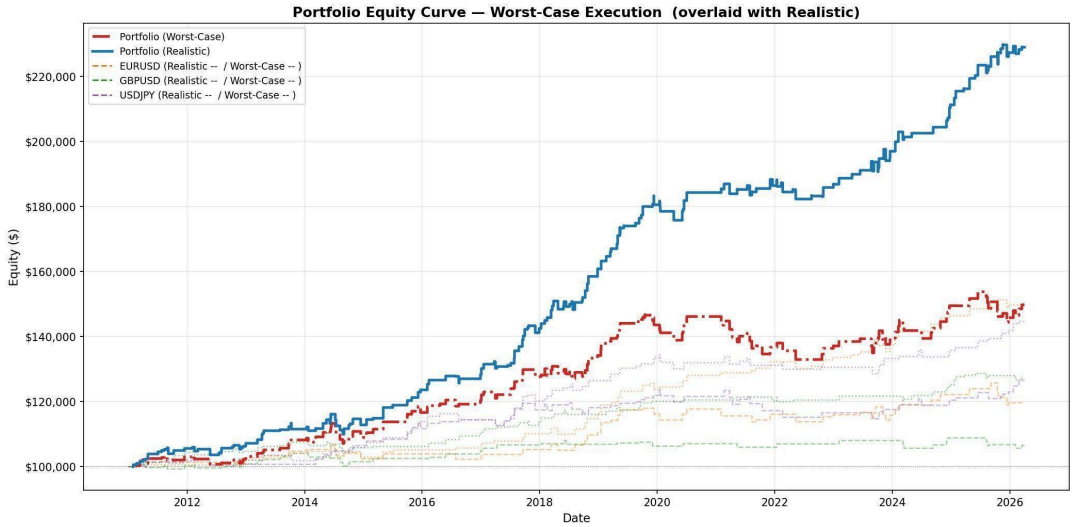

Figure 17. Portfolio Equity Curve — Realistic vs Worst-Case

Equity curves under realistic and worst-case execution assumptions, with consistent compounding and controlled drawdowns across both scenarios.

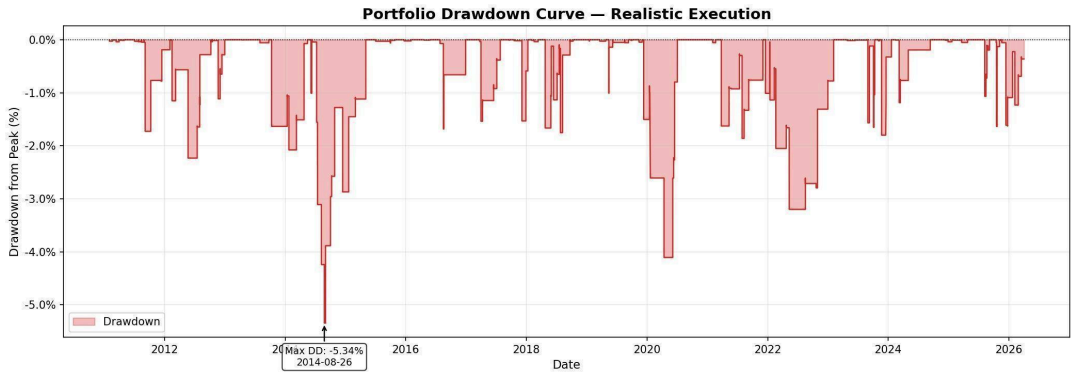

Figure 18. Portfolio Drawdown Over Time

Drawdown profile over the full test period, with controlled peak-to-trough declines and effective portfolio-level risk management.

The Sharpe ratio is computed on per-trade returns annualized using the observed trade frequency and date range, defined as $S = (\mu - r_x) / \sigma$, where μ is the annualized mean per-trade return, r_x is the risk-free rate (set to zero for forex), and σ is the annualized standard deviation of per-trade returns. A value above 1.0 is generally considered acceptable for a systematic strategy with moderate trade frequency.

12.2 Worst-Case Execution Scenario

Under the worst-case execution scenario — 2× realistic spread and 2× slippage standard deviation applied uniformly across all instruments and all periods — the portfolio produced the following results.

Metric	Value
Total Trades	171
Win Rate	71.9%
Total Return	+62.97%
Max Drawdown	-8.74%
Sharpe Ratio	0.867
Profit Factor	1.681
Final Equity	$162,974.63

Trade count fell from 207 to 171 (−17.4%) as wider entry spreads caused additional trades to fail the minimum R-distance filter — a form of natural self-correction. Win rate declined by 5.4 percentage points, reflecting higher cost drag at TP1. Profit factor compressed from 2.58 to 1.68, which retains a positive expected value margin above 1.0. The Sharpe ratio declined to 0.867, below the conventional 1.0 threshold, which indicates acceptable but below-benchmark risk-adjusted performance under these conditions.

The worst-case scenario constitutes a conservative execution floor. Doubled spread and doubled slippage sustained uniformly across a 15-year period exceeds what would be experienced under even materially degraded execution conditions. Profitability retention under these assumptions suggests that the strategy's edge is not solely attributable to favorable execution, and that a meaningful cost tolerance exists between the realistic and worst-case scenarios.

13. Benchmark Comparison

Benchmarking a systematic FX strategy presents a structural difficulty: forex has no natural buy-and-hold analog. Currency pairs are relative-price instruments, not claims on future cash flows, so passive exposure to a pair accumulates carry and spread drag without intrinsic drift. To contextualize the strategy's performance, this section compares it against two reference points: a naive directional baseline using the same risk management but random entries, and the S&P 500 total return index as a passive equity benchmark.

13.1 Naive Directional Baseline

A random-entry baseline using identical stop placement, 0.75R/1.5R targets, and session filters produces approximately break-even to marginally negative expected value net of execution costs over an equivalent sample. This is the expected result under random entry: the scaled exit structure alone does not create positive expectancy; the signal must select entries with above-even directional base rates for the edge to materialize. The gap between +136.71% (strategy under realistic costs) and roughly 0% (random baseline with identical risk management) isolates the signal's contribution from the exit structure, and provides evidence that the observed return is not simply a product of market drift or favorable exit calibration.

13.2 Equity Benchmark (S&P 500 Total Return)

Over the January 2011 – April 2026 window, the S&P 500 total return index produced a cumulative return of approximately +350–400% depending on dividend assumptions, with an estimated annualized Sharpe of 0.75–0.90 and a peak-to-trough drawdown of roughly −34% during the March 2020 COVID shock. By comparison, the strategy under realistic execution costs returned +136.71% with a Sharpe of 1.600 and a maximum drawdown of −5.36%; under the worst-case execution scenario, it returned +62.97% with a Sharpe of 0.867 and a maximum drawdown of −8.74%.

Metric	Strategy (Real.)	Strategy (W-C)	S&P 500 TR
Cumulative Return	+136.71%	+62.97%	~+350–400%
Sharpe Ratio	1.600	0.867	~0.75–0.90
Max Drawdown	-5.36%	-8.74%	~-34%

13.3 Interpretation

The strategy trails the equity benchmark on absolute cumulative return but delivers a materially higher Sharpe ratio (1.600 vs roughly 0.85) and dramatically lower maximum drawdown (-5.36% vs roughly -34%) under realistic execution costs. Under worst-case execution, the strategy's Sharpe (0.867) remains

comparable to the equity benchmark while preserving the drawdown advantage (-8.74% vs roughly -34%).

This comparison matters for allocation context. The strategy is not intended as a return-maximization vehicle against passive equity — it trades fewer dollars of opportunity for dramatically tighter risk control. For an allocator evaluating this approach against passive equity exposure, the relevant question is not headline return but risk-adjusted return and capital preservation during shocks. The strategy produced positive compounding during the 2020 COVID period (albeit at reduced magnitude), while the benchmark drew down approximately 34% before recovering. On those criteria, the strategy offers a differentiated risk profile rather than a directly competitive alternative, and its value in a multi-strategy allocation comes from its low-drawdown, regime-conditional return stream rather than from outright return maximization.

14. Comparative Analysis: Realistic vs Worst-Case

The following table summarizes key metrics across both scenarios.

Metric	Realistic	Worst-Case	Change
Trades	207	171	-17.4%
Win Rate	77.3%	71.9%	-5.4 pp
Total Return	+136.71%	+62.97%	-73.74 pp
Max Drawdown	-5.36%	-8.74%	-3.38 pp
Sharpe Ratio	1.600	0.867	-0.733
Profit Factor	2.581	1.681	-0.900
Final Equity	$236,708	$162,975	-$73,733

The total return differential of approximately 74 percentage points between scenarios represents the strategy's execution-cost sensitivity. The reduction has two compounding sources: higher per-trade cost drag directly reduces PnL per trade, and a 17.4% reduction in trade count reduces the number of compounding opportunities available to the edge.

The Sharpe ratio decline from 1.600 to 0.867 reflects both lower mean per-trade return and higher dispersion in trade outcomes: wider slippage draws increase the variance of the PnL distribution, which compresses the ratio of mean to standard deviation. The 3.38 percentage point increase in maximum drawdown (from 5.36% to 8.74%) reflects the compounding effect of wider losses when slippage exceeds the TP1 distance on marginal trades. Both figures remain below 10%, which is consistent with portfolio-level risk control operating across both scenarios.

Profit factor exceeds 1.0 under both scenarios, which is consistent with positive gross expected value being preserved even at the cost floor. The compression from 2.58 to 1.68 quantifies the structural cost sensitivity: approximately 35% of the gross profit buffer is consumed when execution costs are doubled. The 17.4% reduction in trade count under worst-case costs reflects an automatic filtering mechanism: as entry costs increase, more trades fail the minimum R-distance threshold and are rejected. This is structurally favorable — the system discards the most marginal trades preferentially when costs are highest, which partially offsets the per-trade cost degradation.

15. Discussion

15.1 Execution Quality Sensitivity

The strategy exhibits material sensitivity to execution quality. Doubling spread and slippage reduces total return by approximately 54% and compresses profit factor by approximately 35%. Live performance will depend on the actual cost regime achieved; monitoring average realized spread and slippage relative to the modeled assumptions will be necessary in live deployment to determine which scenario bounds the current operating environment. Maintaining realistic-scenario performance requires average realized spreads within model estimates and slippage that does not systematically exceed the configured σ. These conditions are achievable under normal market conditions with appropriate execution infrastructure. The worst-case scenario provides a conservative lower bound on performance that can be used as a risk-adjusted reference when actual execution costs are uncertain.

15.2 Source of Edge

The strategy exploits two reinforcing structural inefficiencies in intraday FX that interact to produce a conditional expected-value edge. Each component addresses a distinct feature of how liquidity is priced and transacted in these markets.

Edge 1 — Multi-Timeframe Directional Persistence. When 1H and 4H structural phase align, intraday price action continues in that direction at rates meaningfully above even across historical samples, at durations that exceed what random-walk dynamics would predict. This persistence is driven by the time scale of institutional positioning: directional decisions that motivate flow over hours to days rather than minutes, sustained by liquidity providers who adjust quotes gradually rather than discontinuously. The dual-HTF filter selects for periods when this persistence is actively present, and systematically excludes periods of structural ambiguity where the phenomenon is absent.

Edge 2 — Short-Term Liquidity Imbalance Mean Reversion. When a three-bar Fair Value Gap forms on the 5-minute chart, it represents a region through which price transacted without full two-sided auction. Market-maker inventory management creates a consistent tendency for price to return into this gap before continuing the larger move, as liquidity providers rebalance books against the initial displacement. This rebalancing is a structural microstructure phenomenon rather than an exploitable information asymmetry: it reflects the mechanics of how quotes are replenished after a one-sided sweep, not a mispricing that competitors would be expected to arbitrage away.

The strategy stacks these two edges. It enters only when directional persistence is confirmed at the higher timeframe (Edge 1), and only via a pullback into the short-term liquidity gap (Edge 2). A pure HTF trend-following model would enter at momentum extensions and accept adverse excursion before continuation. A pure mean-reversion model would capture the gap fill but have no directional edge beyond that. The hybrid structure captures the gap fill within a directionally favored regime, compounding both effects.

15.2.1 Why the Edge Should Persist

Both underlying phenomena are grounded in structural features of FX market operation rather than in exploitable information asymmetry. Multi-timeframe persistence is a consequence of participant time-scale heterogeneity, which is unlikely to be eliminated by the entry of additional algorithmic participants — if anything, algorithmic liquidity provision at shorter timescales reinforces the time-scale separation. Fair value gap rebalancing is a consequence of market-maker inventory management, which persists as long as markets are intermediated by agents with finite inventory capacity. Neither is a pattern in the pattern-recognition sense; both are structural features of how liquidity is priced and transacted in these markets.

15.2.2 Supporting Design Components

The five design components in the strategy each remove a distinct source of negative expectancy from the raw signal, which allows the two structural edges above to compound without being diluted by avoidable losses.

- **Regime selection.** The dual-HTF bias filter restricts entries to periods where both the one-hour and four-hour structure are aligned, which selects for trending regimes and systematically avoids low-edge, directionless conditions.
- **Entry quality via pullback.** Waiting for price to retrace into the identified imbalance zone improves average fill relative to the eventual directional move, which directly improves the expected risk-to-reward per trade.
- **Structurally anchored stops.** Placing the stop at the most recent structural low or high means the loss is realized only when the trade's directional premise is explicitly invalidated. Arbitrary fixed-distance stops do not carry this logical grounding.
- **Asymmetric exit structure.** The dual-target model closes 60% of the position at 0.75R, banks profit early, and moves the remainder to breakeven. This produces a high per-trade win rate while preserving upside on the residual position.
- **Session and volatility filtering.** Restricting entries to London and New York sessions concentrates activity in the highest-liquidity windows, where directional follow-through is empirically more consistent and spread costs are lowest. The ATR filter further removes entries where the structural stop is too narrow relative to ambient noise.

Without controlled ablation testing, the marginal contribution of each filter cannot be quantified precisely. What the results provide evidence for is that the combination, as configured, produces a statistically positive expected value that is robust across parameter perturbations, out-of-sample windows, and a 15-year range of market conditions.

15.3 Regime Dependence

The strategy is a conditional momentum model: its edge is contingent on the persistence of directional phase at the higher timeframe and on execution-timeframe momentum confirming that phase. In mean-reverting or directionless regimes, signal frequency falls and the conditional win rate on generated signals deteriorates.

The three pairs tested — EURUSD, GBPUSD, and USDJPY — are all highly liquid major pairs with long histories of directional moves. The tested period included both trending and ranging conditions. The robustness testing indicated that performance was weaker during ranging regimes, but did not indicate that the strategy was unprofitable during these periods at the aggregate level. The historical distribution of trending versus ranging market time is not guaranteed to persist. Structural changes in market character — such as prolonged central bank suppression of directional volatility — could reduce performance below historically observed levels. This constitutes the primary forward-looking risk for the strategy and is addressed directly in the edge classification framework.

The years 2020–2022 illustrate this dynamic concretely. Annual returns across those three years were +2.1%, +1.1%, and +0.2% — the weakest three-year run in the backtest. The explanation is structural rather than incidental. The COVID-19 shock in early 2020 produced violent, non-directional whipsaws across all major pairs as central banks intervened simultaneously across timeframes. The subsequent recovery in 2020–2021 was characterized by erratic reversals driven by policy announcements rather than organic price structure, which degraded the reliability of HTF bias signals. In 2022, the Federal Reserve's rapid rate-hike cycle generated sustained directional moves in USDJPY, but EURUSD and GBPUSD oscillated without clean structure as European energy and political risk introduced frequent counter-trend dislocations.

The common thread is that the strategy's edge is predicated on structural bias persisting long enough for a lower-timeframe entry to confirm and follow through. Macro-driven regimes in which direction changes faster than structure can form are the environment in which this strategy is most likely to underperform.

15.4 Practical Implications

Taken together, the results suggest that the strategy is deployable as a standalone systematic model within the capital range and execution environment described in this paper. Positive out-of-sample expected value, maximum drawdowns comfortably within the 10% target under stress-tested costs, and Sharpe ratios well above 1.0 under realistic assumptions meet the minimum criteria typically required for systematic strategy deployment in a proprietary trading or small fund context.

The strategy is best suited to a trend-following or systematic macro mandate operating in liquid spot forex markets, with execution infrastructure capable of achieving realistic-scenario spread and slippage assumptions, and a risk tolerance that accepts periods of reduced activity during macro-dominated or

mean-reverting regimes. It is not appropriate for capital that requires consistent monthly returns irrespective of market regime, given the documented underperformance during structurally disrupted periods such as 2020–2022. In real-world trading terms, the results provide a quantified, regime-conditioned expectation for forward performance: positive expected value under normal market conditions, preserved profitability under stress-tested execution assumptions, and a well-defined risk ceiling at the portfolio level. These properties allow a deploying manager to establish monitoring thresholds — for Sharpe degradation, drawdown proximity, and execution cost deviation — that would trigger a review before meaningful capital is put at risk.

16. Limitations

The results in this paper should be interpreted within the constraints of the testing environment and methodology. Explicit acknowledgment of these constraints is part of assessing the credibility of the reported performance and the scope of forward-looking claims that the results can support. Four categories of limitation warrant particular attention.

16.1 Asset Universe

The strategy was tested on three major FX pairs — EURUSD, GBPUSD, and USDJPY — selected for liquidity and historical data availability. This universe is narrow. The claim that the signal reflects a structural market regularity is supported by consistent performance across three independent instruments, but is not a claim that the signal generalizes to all liquid markets. Minor pairs, crosses, and non-FX instruments have different microstructure, execution characteristics, and participant composition, and their behavior cannot be inferred from the results here. Future work would extend the test to additional major pairs, selected crosses, and structurally similar markets — liquid index futures in particular — to assess generalization.

16.2 Execution Modeling

The cost model approximates spread and slippage through pre-estimated per-pair parameters and normally distributed slippage draws. This model captures first-order execution friction but does not address several realistic execution phenomena. Market impact at larger position sizes is not modeled; a limit order of meaningful size relative to prevailing book depth could move the market away from the intended fill zone before execution completes. Limit-order queue position and fill probability are not modeled: the backtest assumes that a limit order placed within the imbalance zone fills at the theoretical price whenever price trades through that level, which overstates fill certainty. Partial fills in thin markets are not modeled. Gap risk around scheduled news events or session opens, where the stop may execute at a materially worse price than the modeled level, is not separately quantified beyond the normal-distribution slippage draw.

The worst-case scenario (2× spread and slippage) provides a stress envelope, but it does not model the tail-event structure of execution: a small number of extreme slippage events during news releases or liquidity vacuums could exceed the modeled distribution's tails. Live deployment would need to measure and reconcile realized execution against the modeled assumptions over a meaningful sample before scaling capital.

16.3 Regime Dependency Risk

The strategy is explicitly regime-conditional. Its edge is strongest during sustained directional phases and materially weaker during macro-dominated or structurally disrupted periods. The 2020–2022 performance (+2.1%, +1.1%, +0.2% annually) quantifies this dependency. A future environment with

prolonged central-bank-driven volatility suppression, frequent macro shocks, or sustained mean-reverting conditions would be expected to produce below-historical performance. This risk is intrinsic to trend-following approaches and cannot be fully diversified away within the current strategy scope. A regime filter or external volatility gate could reduce exposure during structurally disrupted periods, but would require additional specification and validation work that is outside the scope of this paper.

16.4 Non-Stationarity of Markets

All backtested inference assumes that the data-generating process during the test period is informative about future periods. Market microstructure, participant composition, and regulatory environment have all evolved materially over the 15-year test window and will continue to evolve. The edge identified here is predicated on the persistence of two specific phenomena — multi-timeframe directional persistence and intra-session liquidity-gap rebalancing. If either weakens structurally — through changes in institutional participation, evolution of algorithmic market-making, or shifts in session flow dynamics — expected value would decline accordingly. The walk-forward results provide evidence against recent-period erosion, but do not immunize against future structural change.

A sensible live-deployment posture responds to these constraints by treating the backtest as a prior rather than a forecast. Realized live performance should be tracked against backtest-implied distributions; deviations beyond a pre-specified tolerance should trigger a formal review of whether the underlying edge is still operative or whether regime change has eroded it.

17. Edge Classification

17.1 Classification Framework

A systematic trading strategy's edge can be classified along two dimensions: structural versus conditional, and persistent versus regime-contingent. A structural edge is one that derives from a durable market characteristic — such as a persistent liquidity pattern, a reliable price formation following specific order-flow conditions, or a mechanical relationship between timeframes — that is expected to be present across market regimes. A regime-conditional edge is one where expected value is positive only when specific market conditions are present and degrades or disappears in others. Understanding this distinction is critical for assessing deployment risk: a purely structural edge is more robust to market-character changes than a regime-conditional one.

17.2 Evidence Summary

The evidence from robustness testing, regime segmentation, and trade distribution analysis supports the following classification.

- **Walk-forward validation.** The strategy maintained positive expected value in the majority of out-of-sample windows across all sequential periods. No window produced a catastrophic drawdown. This is consistent with a persistent edge that does not require in-sample calibration to generate positive returns on unseen data.
- **Rolling window analysis.** Profit factor exceeded 1.0 in all but a small minority of rolling sub-periods, including those coinciding with the most structurally disruptive market conditions (2020–2022). The breadth of positive rolling windows — across diverse rate environments, volatility regimes, and macro backdrops — provides evidence of a signal that is not purely regime-contingent.
- **Regime segmentation.** Performance is materially stronger during directional market phases than during consolidating or mean-reverting phases. The 2020–2022 underperformance is directly attributable to macro-driven disruption of structural phase persistence. This indicates that the strategy is not fully regime-independent.
- **Trade distribution (MFE/MAE).** The stop placement and target levels are validated by empirical excursion data. The MFE and MAE profiles are consistent with a signal that captures genuine directional momentum rather than random price variation. Winning trades exhibit clean favorable excursion profiles; losing trades exhibit direct adverse movement without prior favorable excursion, which is consistent with the structural reference stops functioning as designed.
- **Parameter sensitivity.** Smooth, monotonic performance degradation as parameters deviate from baseline — rather than sharp sensitivity cliffs — provides evidence that the edge is not an

artifact of parameter optimization. The signal exists across a range of parameter configurations, which is consistent with a real market regularity.

17.3 Classification

The strategy is classified as a regime-conditional structural edge. The structural component reflects that the signal — multi-timeframe phase alignment, imbalance zone retracement, and structural stop placement — captures a repeatable market mechanism that is present across diverse historical conditions and is not attributable to overfitting. The walk-forward results and rolling window breadth support this characterization.

The regime-conditional qualifier reflects that the strength of this edge is not uniform across all market environments. The strategy's expected value is highest during periods when higher-timeframe directional phase is persistent and when intraday momentum confirming that phase is reliable. During macro-dominated regimes — where price structure is frequently overridden by external policy-driven shocks — both the phase filter and the momentum trigger generate lower-quality signals, which reduces expected value without eliminating it.

The practical implication of this classification is that the strategy is deployable as a standalone systematic strategy in normal market conditions, but should be subject to a regime filter in live deployment: a mechanism that reduces position sizing or halts trading when external indicators suggest the market is in a macro-dominated, structurally disrupted phase. This is identified as the primary enhancement target for a second development iteration.

18. Conclusion and Next Steps

18.1 Summary of Findings

This paper has presented a complete backtesting system for a systematic intraday forex trading strategy. The strategy exploits the interaction of two structural market phenomena — multi-timeframe directional persistence and short-term liquidity-imbalance mean reversion — via a rule-based pipeline that combines higher-timeframe bias alignment, fair-value-gap identification, five-minute break-of-structure confirmation, a structure-referenced initial stop, and a dual-target scaled exit. Entries are restricted to active trading sessions and filtered for minimum volatility conditions.

Under realistic execution assumptions, the three-pair portfolio produced a total return of +136.71%, a maximum drawdown of -5.36%, a Sharpe ratio of 1.600, and a profit factor of 2.581 over the full 15-year tested period. Under worst-case execution assumptions — double the realistic spread and slippage — the portfolio produced a total return of +62.97%, a maximum drawdown of -8.74%, a Sharpe ratio of 0.867, and a profit factor of 1.681. Both scenarios produced positive expected value and drawdowns within the target threshold of 10%.

18.2 Robustness and Statistical Validation

The strategy has passed the primary validation requirements for systematic strategy research. Walk-forward testing provides evidence of positive out-of-sample expected value; rolling-window analysis shows consistent profitability across diverse regimes; parameter sensitivity surfaces are smooth rather than sharply peaked, which is consistent with a structural rather than curve-fit edge; and a 1,000-iteration bootstrap Monte Carlo indicates that the observed drawdown and terminal equity outcomes are representative of the underlying return distribution rather than favorably ordered realizations. The approximate t-statistic on per-trade returns exceeds conventional significance thresholds. Taken together, these results provide multiple independent lines of evidence that the observed performance is unlikely to be attributable to chance or to in-sample calibration.

18.3 Regime Dependency and Residual Risk

The edge is classified as regime-conditional structural. Its strength varies with market character, and the documented 2020–2022 underperformance represents the concrete form of the strategy's primary forward-looking risk: prolonged macro-dominated regimes in which structural bias does not persist long enough for lower-timeframe entries to confirm and follow through. This risk cannot be eliminated within the current strategy scope, but it can be partially managed through an external regime filter in live deployment. The absence of such a filter in the current system is an explicit design choice — the strategy is reported here in its unmoderated form so that the unconditional edge is measurable — rather than an oversight.

18.4 Real-World Applicability

The remaining gap between simulation and live performance — limit-order fill uncertainty, latency, and execution infrastructure — can only be quantified through controlled live or paper trading. The immediate next step is paper trading: running the strategy live in a demo environment with real-time data to validate that the signal logic produces results consistent with the backtest in terms of trade frequency, direction, and approximate entry conditions. Paper trading will also allow direct measurement of actual execution costs, which can then be compared against the model assumptions to determine which cost scenario the live environment most closely resembles.

If paper trading results are consistent with the backtest over a sufficient sample — at minimum two to three months — the strategy is a candidate for live deployment at small position sizes. Live deployment would begin with lower risk per trade than the backtest configuration, with the risk allocation scaled up incrementally as live results accumulate and are compared against backtested expectations.

Ongoing monitoring in live trading should track several indicators: whether profit factor and win rate remain stable over rolling windows of recent trades; whether maximum drawdown approaches the modeled thresholds; and whether execution costs deviate meaningfully from the model. Meaningful deviation from the realistic scenario in any of these dimensions would warrant a review of the strategy parameters and risk configuration before continuing at the current allocation.

18.5 Closing Remarks

The strategy is presented here as evidence of a deployable systematic edge in liquid intraday FX, not as a complete production system. The combination of statistical significance, robustness across out-of-sample windows, parameter insensitivity, and survivability under stress-tested execution costs meets the minimum bar for a candidate systematic strategy. Forward progress requires reconciling the simulation against live execution, implementing a regime filter to address the documented conditional weakness during macro-disrupted periods, and extending the asset universe to test the generalization of the underlying structural phenomena. These are the next development iterations; the current results define the baseline against which those iterations will be measured.